

STM32F402RE

Drivers

Bare-metal peripheral drivers for the STM32F401RE microcontroller, developed from scratch using reference manual.

Overview

This project provides low-level drivers for the STM32F401RE ARM Cortex-M4 microcontroller. The drivers are written in C and provide a hardware abstraction layer for various peripherals.

Supported Peripherals

Peripheral	Description
GPIO	General Purpose Input/Output
SPI	Serial Peripheral Interface
I2C	Inter-Integrated Circuit
USART	Universal Synchronous/Asynchronous Receiver/Transmitter
RCC	Reset and Clock Control

Project Structure

stm32f401_project/

```
| Drivers/
```

```
| | — inc/                                # Driver header files
```

```
| | | — stm32f401xx.h # MCU specific header
```

```
| | | — stm32f401xx_gpio_driver.h # GPIO driver header
```

```
└─ stm32f401xx spi driver.h  # SPI driver header
```

```
└─ stm32f401xx i2c driver.h    # SPI driver header
```

```
| |   └─ stm32f401xx_rcc_driver.h   # SPI driver header
```

```
| |   └─ stm32f401xx usart driver.h # SPI driver header
```

```
|   └─ src/                                # Driver source files
```

```
| | | — stm32f401xx gpio driver.c
```

```
| |      └─ stm32f401xx spi driver.c
```

```
└─ stm32f401xx i2c driver.c
```

```
└─ stm32f401xx rcc driver.c
```

```
| | └─ stm32f401xx usart driver.c
```

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```
|_ Core/
```

```
| | └─ inc/
```

```
|   └─ src/                                # Application source files
```

```
| |— Startup                                # Startup assembly file
```

— 100 —

```
|— Example Tasks                                # driver applications
```

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```
|— Services # printf, usart debugging, delay
```

Hardware

- MCU: STM32F401RE (ARM Cortex-M4, 84 MHz)
- Board: NUCLEO-F401RE
- Flash: Up to 512 KB
- SRAM: 96 KB

Vedio link: [driver demo videos](#)